Project Python Foundations: FoodHub Data Analysis

Marks: 60

Context

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows the restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin of the delivery order from the restaurants.

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. They want to analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience. Suppose you are hired as a Data Scientist in this company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company to improve the business.

Data Description

The data contains the different data related to a food order. The detailed data dictionary is given below.

Data Dictionary

- order_id: Unique ID of the order
- customer_id: ID of the customer who ordered the food
- restaurant_name: Name of the restaurant
- cuisine_type: Cuisine ordered by the customer
- cost: Cost of the order
- day_of_the_week: Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)
- rating: Rating given by the customer out of 5
- food_preparation_time: Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.
- delivery_time: Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information.

Let us start by importing the required libraries

In [1]:
```
# import libraries for data manipulation
import numpy as np
import pandas as pd

# import libraries for data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

In [35]:
```
!jupyter nbconvert --to html
```

```
[NbConvertApp] Converting notebook /content/PYF_Project_LearnerNotebook_Sept_28_Siddiq.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 19 image(s).
[NbConvertApp] Writing 1347498 bytes to /content/PYF_Project_LearnerNotebook_Sept_28_Siddiq.html
```

Understanding the structure of the data

In [4]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

In [3]:
```
# read the data
df = pd.read_csv('/content/foodhub_order.csv')
# returns the first 5 rows
df.head()
```

Out[3]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24

Observations:

The DataFrame has 9 columns as mentioned in the Data Dictionary. Data in each row corresponds to the order placed by a customer.

Question 1: How many rows and columns are present in the data? [0.5 mark]

In [7]:
```
print("Number of rows: ", df.shape[0])
print("Number of columns: ", df.shape[1])
```

```
Number of rows:  1898
Number of columns:  9
```

Observations:

There are 1898 rows in the data and there are 9 columns in total for this data, wherein each row represents each order.

Question 2: What are the datatypes of the different columns in the dataset? (The info() function can be used) [0.5 mark]

In [8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   order_id               1898 non-null   int64  
 1   customer_id            1898 non-null   int64  
 2   restaurant_name        1898 non-null   object 
 3   cuisine_type           1898 non-null   object 
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object 
 6   rating                 1898 non-null   object 
 7   food_preparation_time  1898 non-null   int64  
 8   delivery_time          1898 non-null   int64  
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

Observations:

There are 5 numerical columns, in which 4 are integers and 1 column is a float. There are also 4 columns that are categorical, which has a datatype of object. There are no missing values, and all the Dtype is correct for each column.

Question 3: Are there any missing values in the data? If yes, treat them using an appropriate method. [1 mark]

In [9]:
```
df.isnull().sum()
```

Out[9]:

	0
order_id	0
customer_id	0
restaurant_name	0
cuisine_type	0
cost_of_the_order	0
day_of_the_week	0
rating	0
food_preparation_time	0
delivery_time	0

dtype: int64

Observations:

As seen from the table above, there are no missing values from the DataFrame.

Question 4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed? [2 marks]

In [5]:
```
df.describe().round(2).T
```

Out[5]:

	count	mean	std	min	25%	50%	75%	max
order_id	1898.0	1477495.50	548.05	1476547.00	1477021.25	1477495.50	1477969.75	1478444.00
customer_id	1898.0	171168.48	113698.14	1311.00	77787.75	128600.00	270525.00	405334.00
cost_of_the_order	1898.0	16.50	7.48	4.47	12.08	14.14	22.30	35.41
food_preparation_time	1898.0	27.37	4.63	20.00	23.00	27.00	31.00	35.00
delivery_time	1898.0	24.16	4.97	15.00	20.00	25.00	28.00	33.00

Observations:

Above is the statistical summary of all the numerical columns from the DataFrame. The minimum time it takes for the food to be prepared is 20 mins. The maximum time it takes for the food to be prepared is 35 mins. The average time it takes for the food to be prepared is about 27 minutes.

Question 5: How many orders are not rated? [1 mark]

In [15]:
```
not_rated = (df['rating'] == 'Not given').sum()
print("Number of orders not rated:", (not_rated))
```

```
Number of orders not rated: 736
```

There are 736 orders that are not rated out of 1898 ratings.

Exploratory Data Analysis (EDA)

Univariate Analysis

Question 6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.) [9 marks]

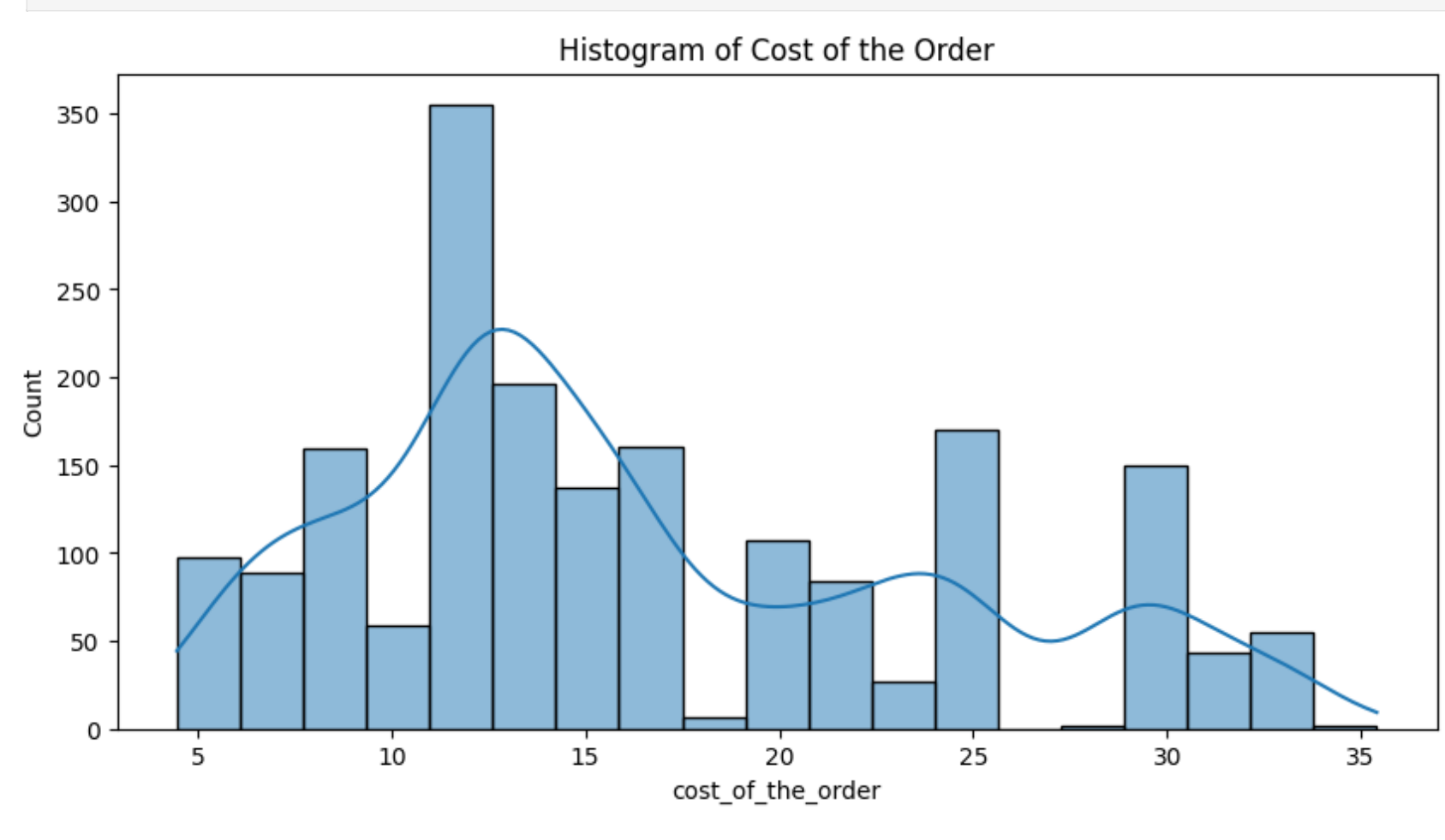
Numerical Columns:

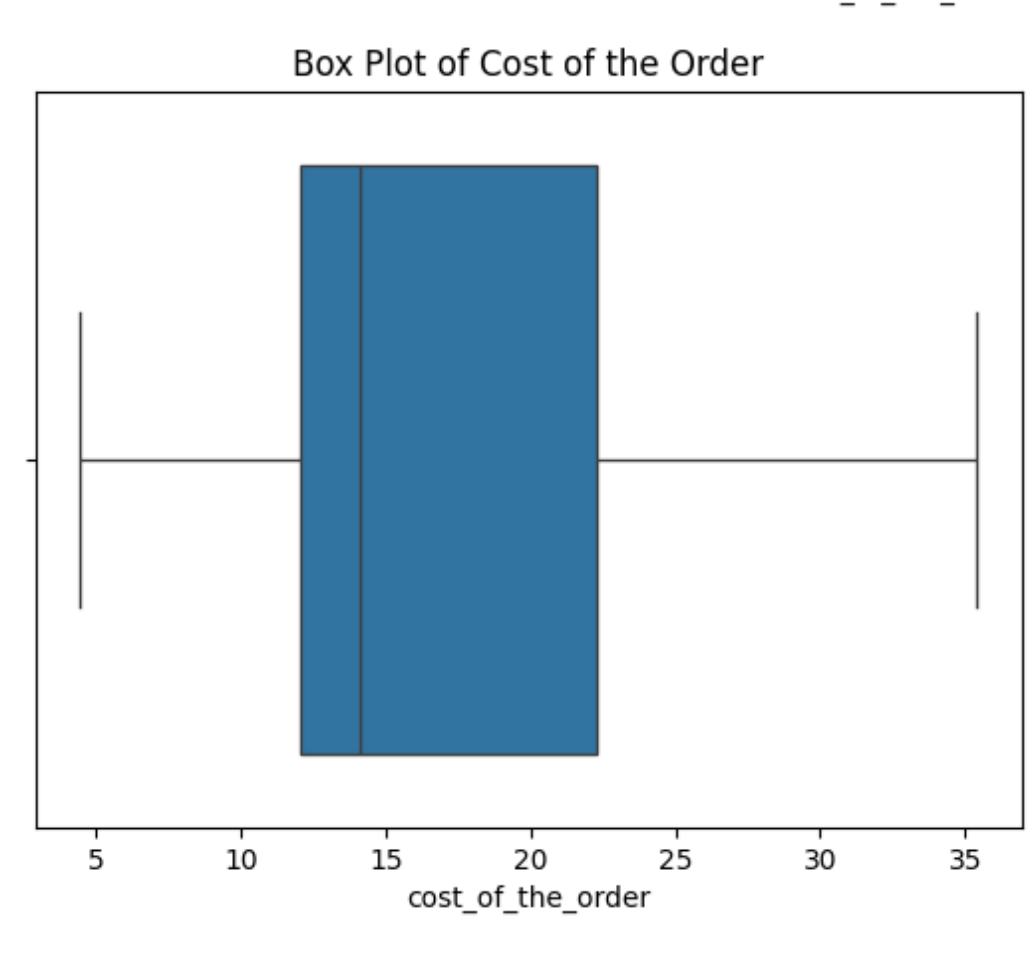
In [13]:
```
plt.figure(figsize=(10, 5))
sns.histplot(df['cost_of_the_order'], kde=True)
plt.title('Histogram of Cost of the Order')

plt.show()

sns.boxplot(x=df['cost_of_the_order'])
plt.title('Box Plot of Cost of the Order')

plt.show()
```





Observations of Cost of Order:

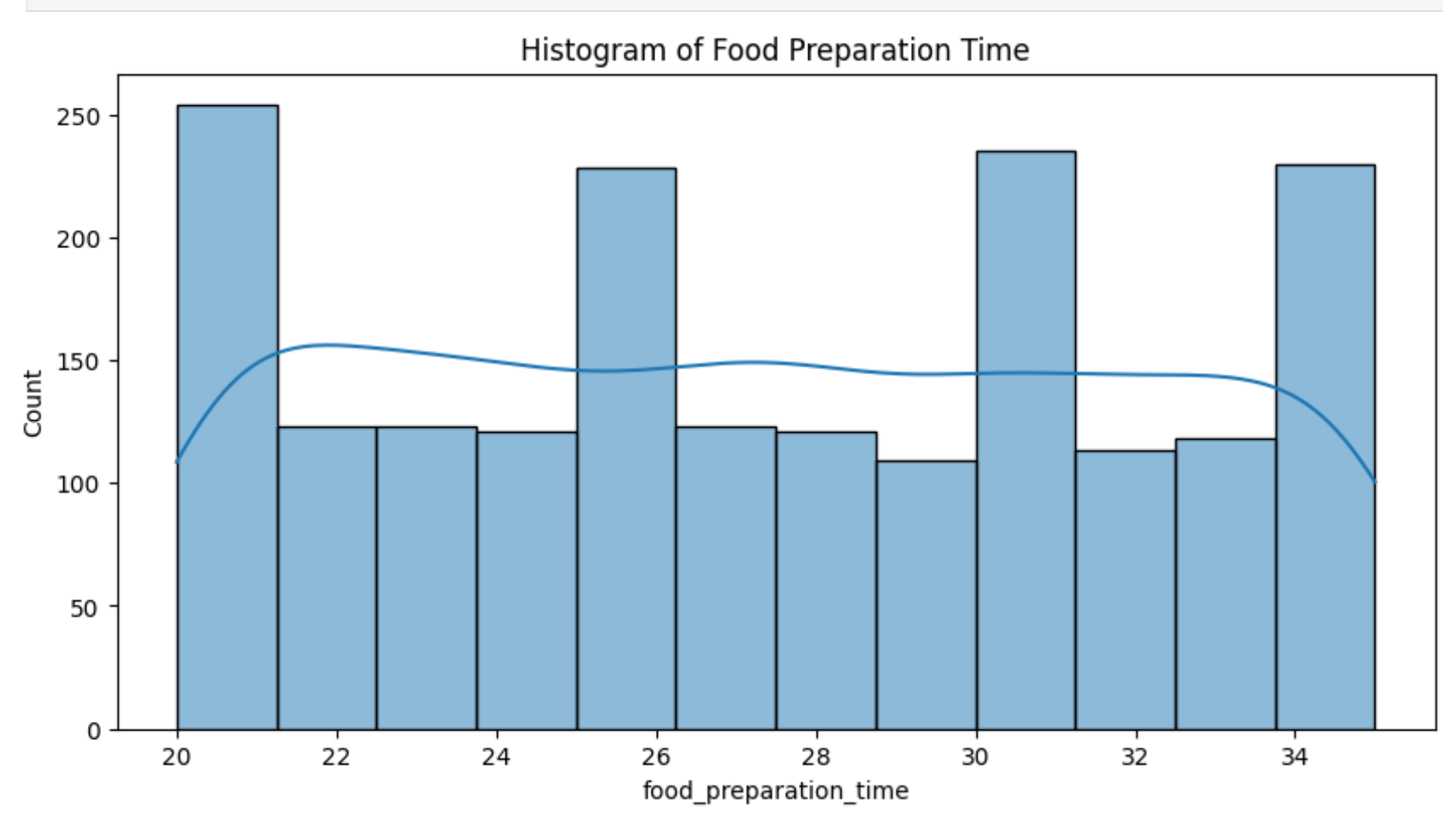
From the Histogram above, it shows that the distribution is slightly right-skewed, meaning most of the orders cost approximately between 10-20 Dollars. From the Box Plot above, the median cost is between 13-14 Dollars. The IQR is approximately between 12-22 Dollars, which means most of the orders cost between that range. From the distribution above, it is also seen that there are orders that cost more around 30-35 Dollars, but it is very rare. Also, there are no outliers, which means all the orders are within the range.

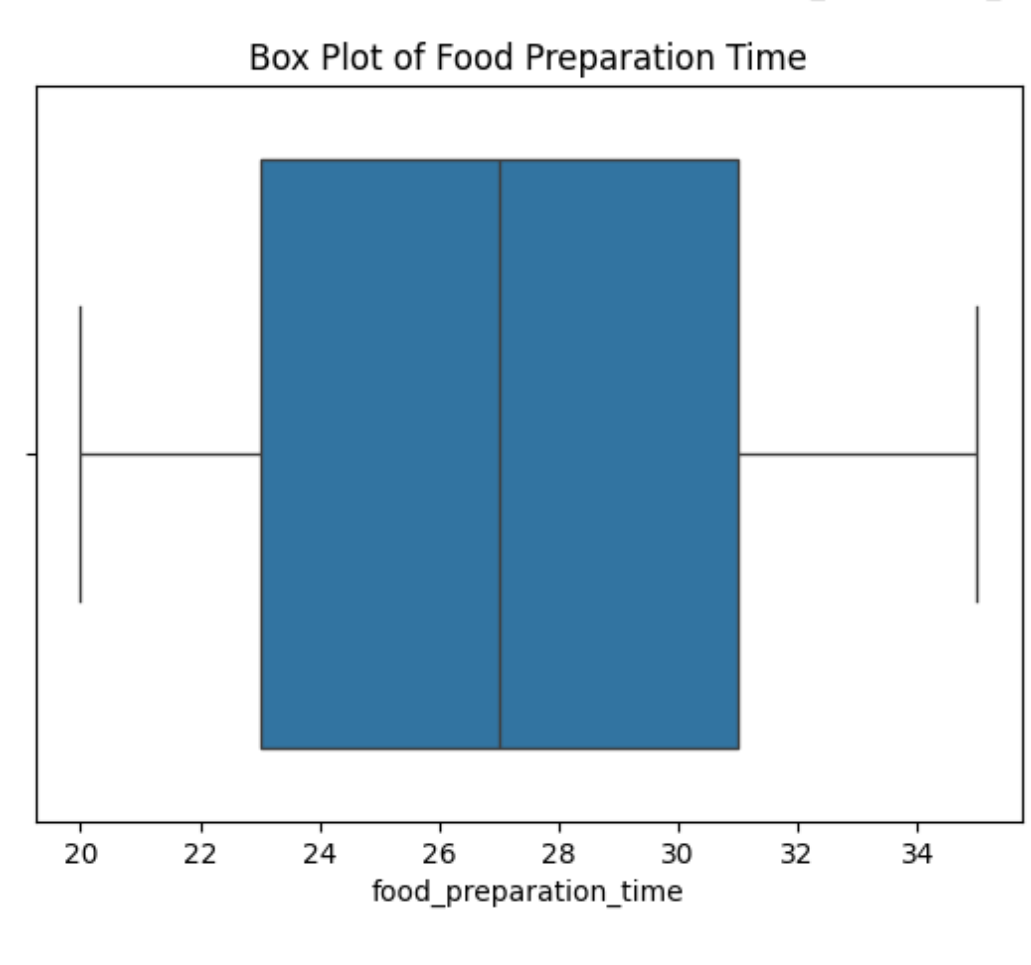
In [14]:
```
plt.figure(figsize=(10, 5))
sns.histplot(df['food_preparation_time'], kde=True)
plt.title('Histogram of Food Preparation Time')

plt.show()

sns.boxplot(x=df['food_preparation_time'])
plt.title('Box Plot of Food Preparation Time')

plt.show()
```





Observations of Food Preparation Time:

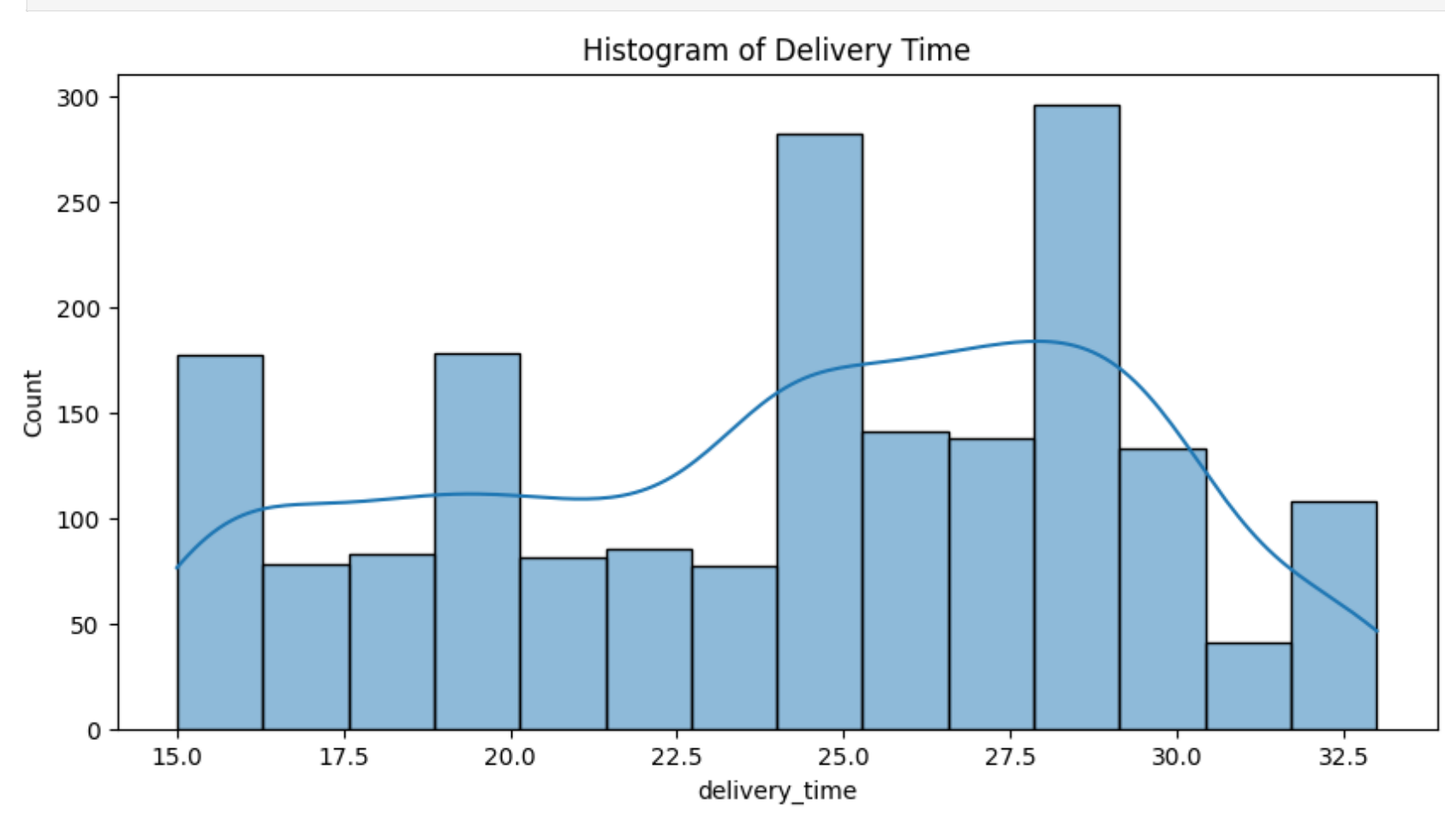
From the histogram above, it shows that it takes from 20 to 35 minutes to prepare food. The distribution looks uniform, without any skewness. From the box plot above, the median time to prepare food is around 27-28 minutes. The IQR is between 23 to 31 minutes, which means most of the orders are prepared within that range. There are no outliers as seen the graph above. It shows that the food preparation time is very consistent and evenly distributed.

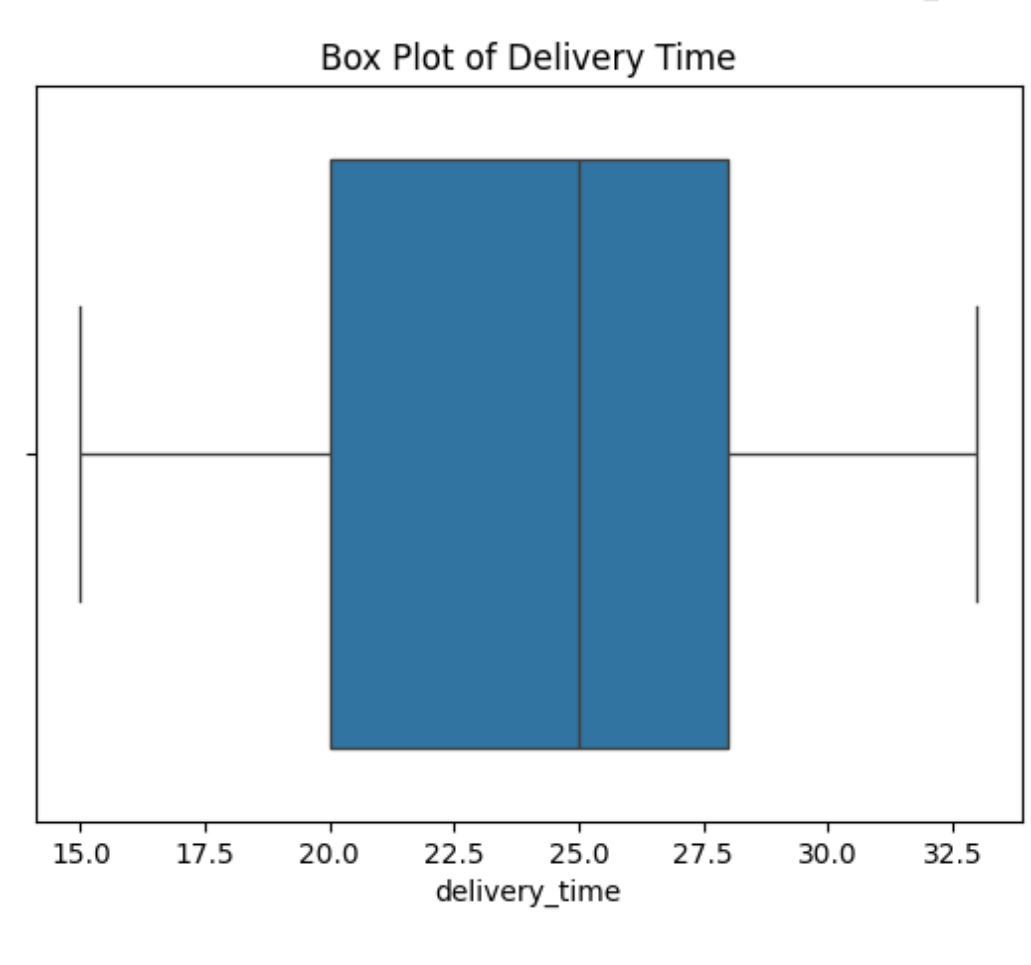
In [6]:
```
plt.figure(figsize=(10, 5))
sns.histplot(df['delivery_time'], kde=True)
plt.title('Histogram of Delivery Time')

plt.show()

sns.boxplot(x=df['delivery_time'])
plt.title('Box Plot of Delivery Time')

plt.show()
```



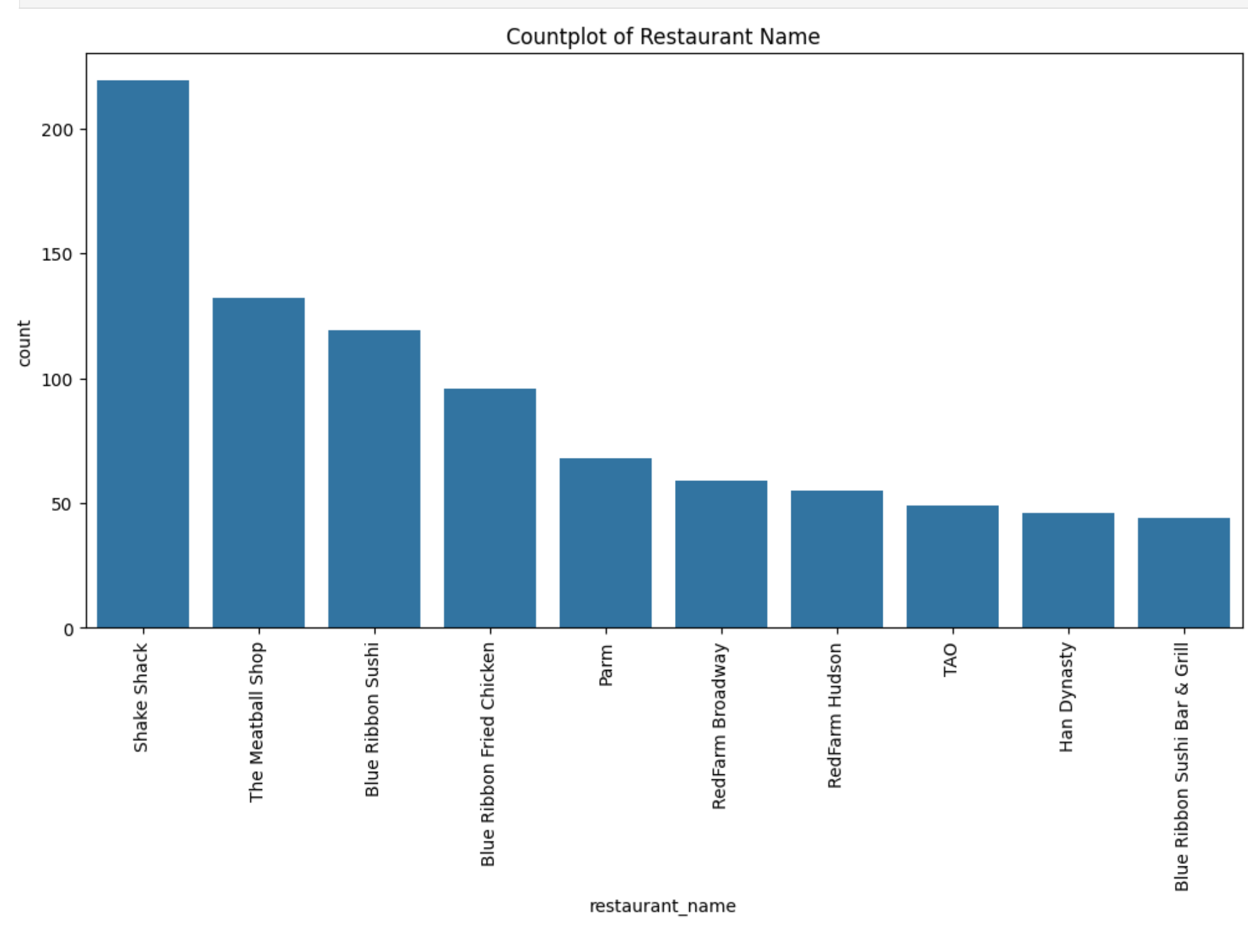


Observations of Delivery Time:

From the histogram above, it shows that it takes about 15 to 33 minutes to deliver food. The distribution peaks around 20,25,28 minutes but is fairly even throughout. From the box plot, it shows that the median time it takes to deliver food is 25 minutes. The IQR is between 20 to 28 minutes, which means most of the delivery orders are delivered within that range. There are no outliers as seen from the box plot above. From the plots above, it takes around 20-28 minutes to deliver food.

Categorical Columns:

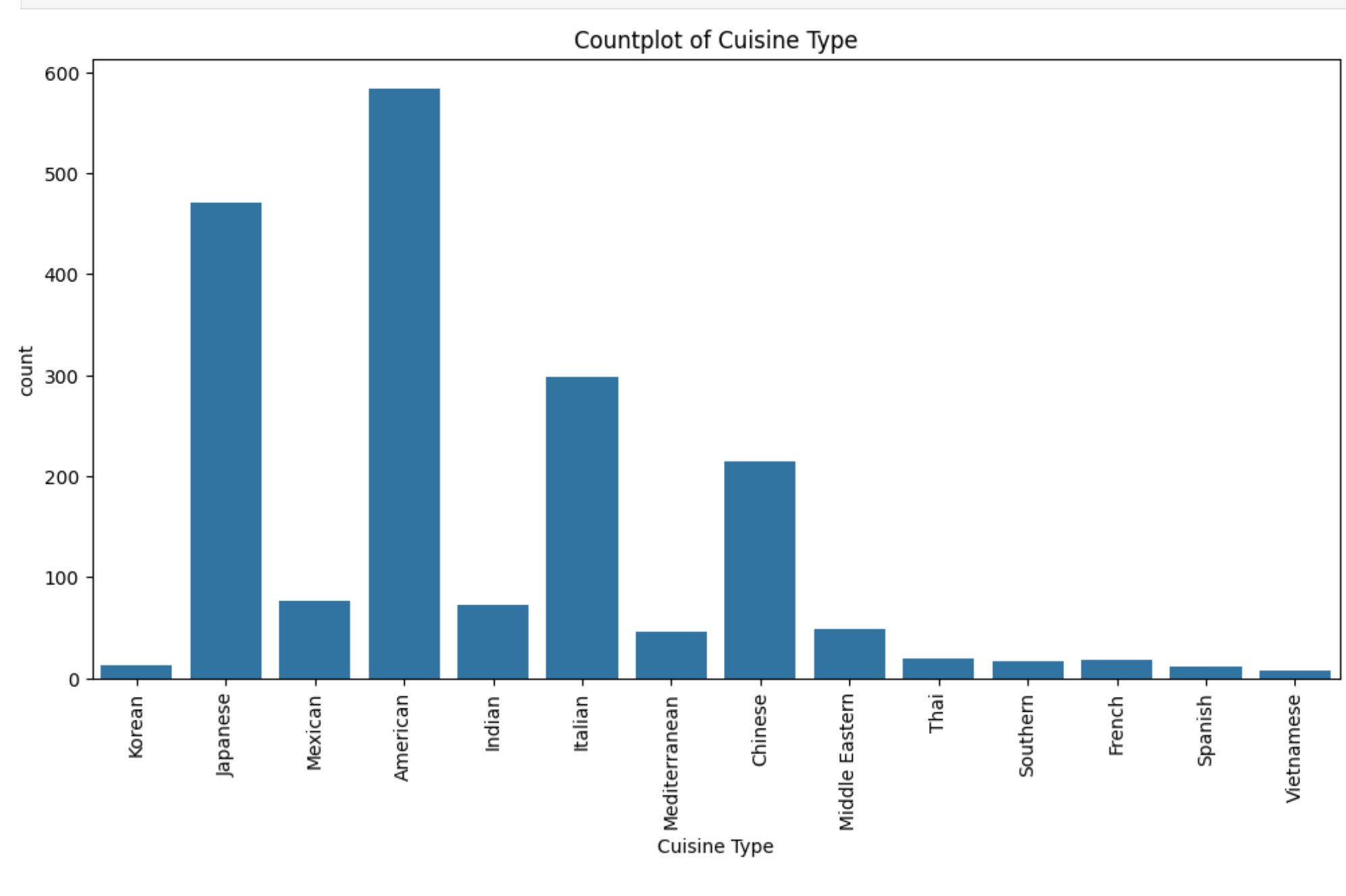
In [25]:
```
plt.figure(figsize =(12, 6))
sns.countplot(data =df, x = 'restaurant_name',
              order = df['restaurant_name'].value_counts().head(10).index)
plt.title('Countplot of Restaurant Name')
plt.xticks(rotation = 90);
```



Observations of restaurant name:

From the countplot seen above, Shake Shack is the most popular restaurant, with the highest number of orders. The Meatball Shop and Blue Ribbon Sushi and also the second and third highest restaurants that customers order from, following by Blue Ribbon Fried Chicken with good number of orders.

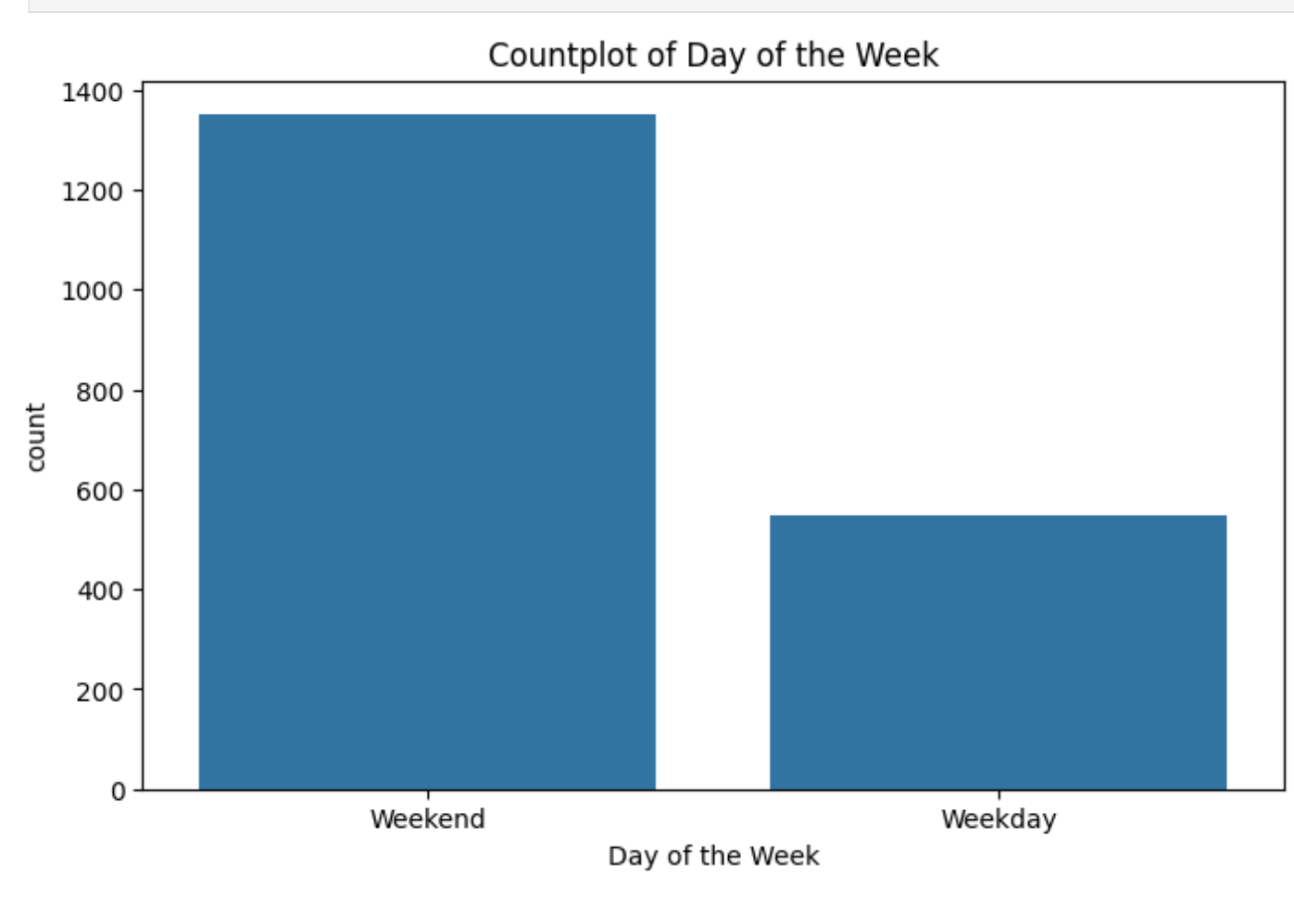
In [16]:
```
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='cuisine_type')
plt.title('Countplot of Cuisine Type')
plt.xlabel('Cuisine Type')
plt.xticks(rotation=90)
plt.show()
```



Observations of Cuisine Type:

From the countplot above, the highest orders are from American cuisine, around 580 orders and looks the most popular among customers. The second most popular is the Japanese cuisine with around 470 orders. The third popular are Italian and Chinese cuisines, and other cuisines are relatively lower.

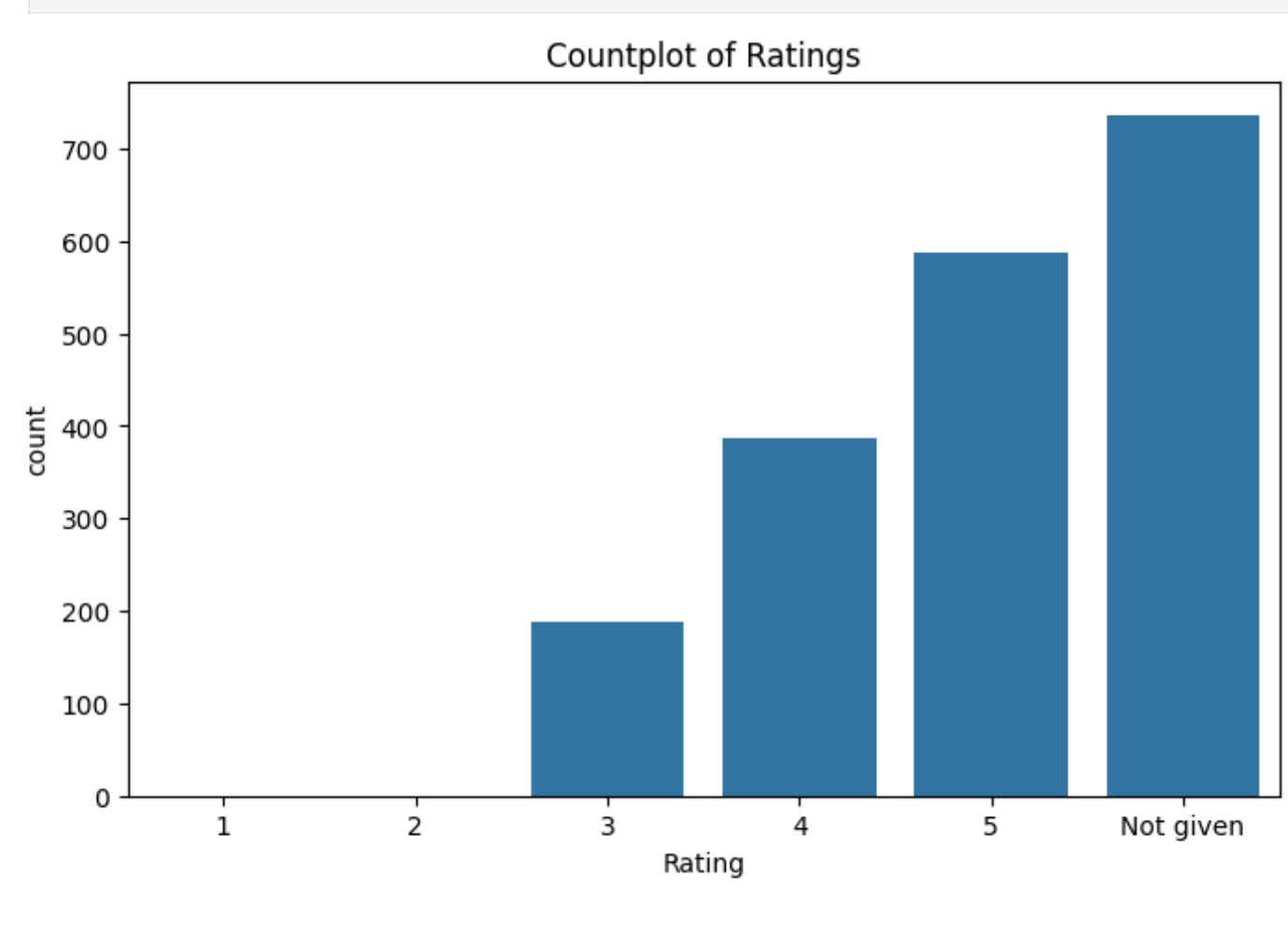
In [8]:
```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='day_of_the_week')
plt.title('Countplot of Day of the Week')
plt.xlabel('Day of the Week')
plt.show()
```



Observations of Day of the Week:

From the countplot seen above, it shows that there are more orders on the weekends than the weekdays, and is almost double, by weekend orders are around 1350 and weekday orders are only about 550.

In [10]:
```
plt.figure(figsize=(8, 5))
order = [1, 2, 3, 4, 5, 'Not given']
sns.countplot(data=df, x='rating', order=order)
plt.title('Countplot of Ratings')
plt.xlabel('Rating')
plt.show()
```



Observations of Rating:

From the countplot seen above, it looks like most of the ratings are missing, as most of the customers didn't provide a rating. Among the ratings, most of the customers rated a 5, leaving a positive feedback. The other ratings are 3 and 4, however, no one ever rated it 1 or 2, which is a good sign. The business does have positively skewed rating, however, there are large number of missing ratings as well.

Question 7: Which are the top 5 restaurants in terms of the number of orders received? [1 mark]

In [11]:
```
df['restaurant_name'].value_counts().head()
```

Out[11]:

restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

dtype: int64

Observations:

Top 5 restaurants by number of orders received are Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken, and Parm.

Question 8: Which is the most popular cuisine on weekends? [1 mark]

In [9]:
```
weekend_orders = df[df['day_of_the_week'] == 'Weekend']
popular_cuisine_weekends = weekend_orders['cuisine_type'].value_counts().idxmax()

print("The most popular cuisine on weekends is:", popular_cuisine_weekends)
```

```
The most popular cuisine on weekends is: American
```

Observations:

The most popular cuisine on weekends is American cuisine.

Question 9: What percentage of the orders cost more than 20 dollars? [2 marks]

In [12]:
```
percentage_above_20 = round(100*len(df[df['cost_of_the_order'] > 20])/len(df),2)

print(f"Percentage of orders costing more than $20: {percentage_above_20:.2f}%")
```

```
Percentage of orders costing more than $20: 29.24%
```

Observations:

29.24% of orders cost more than 20 dollars.

Question 10: What is the mean order delivery time? [1 mark]

In [31]:
```
mean_delivery_time = df['delivery_time'].mean()
print(f"Mean order delivery time: {mean_delivery_time:.2f} minutes")
```

```
Mean order delivery time: 24.16 minutes
```

Observations:

The average delivery time of the order is around 24 minutes.

Question 11: The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed. [1 mark]

In [13]:
```
df['customer_id'].value_counts().head(5)
```

Out[13]:

customer_id	count
52832	13
47440	10
83287	9
250494	8
259341	7

dtype: int64

Observations:

As seen above, top 5 most frequent customers and their number of orders they placed are:

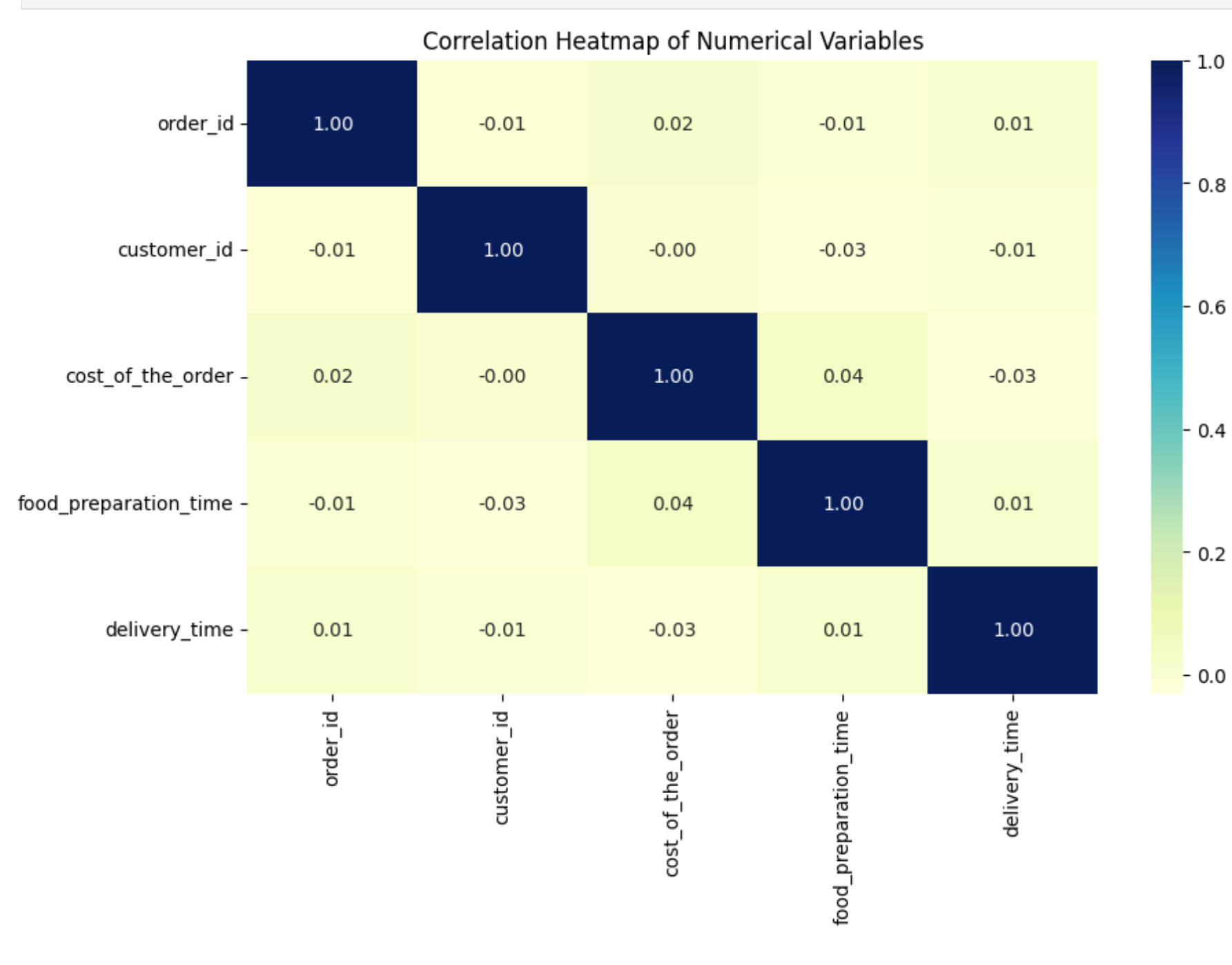
Customer ID: 52832 with 13 orders; Customer ID: 47440 with 10 orders; Customer ID: 83287 with 9 orders; Customer ID: 250494 with 8 orders; Customer ID: 259341 with 7 orders.

Multivariate Analysis

Question 12: Perform a multivariate analysis to explore relationships between the important variables in the dataset. (It is a good idea to explore relations between numerical variables as well as relations between numerical and categorical variables) [10 marks]

Heatmap for correlations of all numerical columns

In [56]:
```
plt.figure(figsize=(10, 6))
sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, fmt=".2f", cmap="YlGnBu")
plt.title('Correlation Heatmap of Numerical Variables')
plt.show()
```

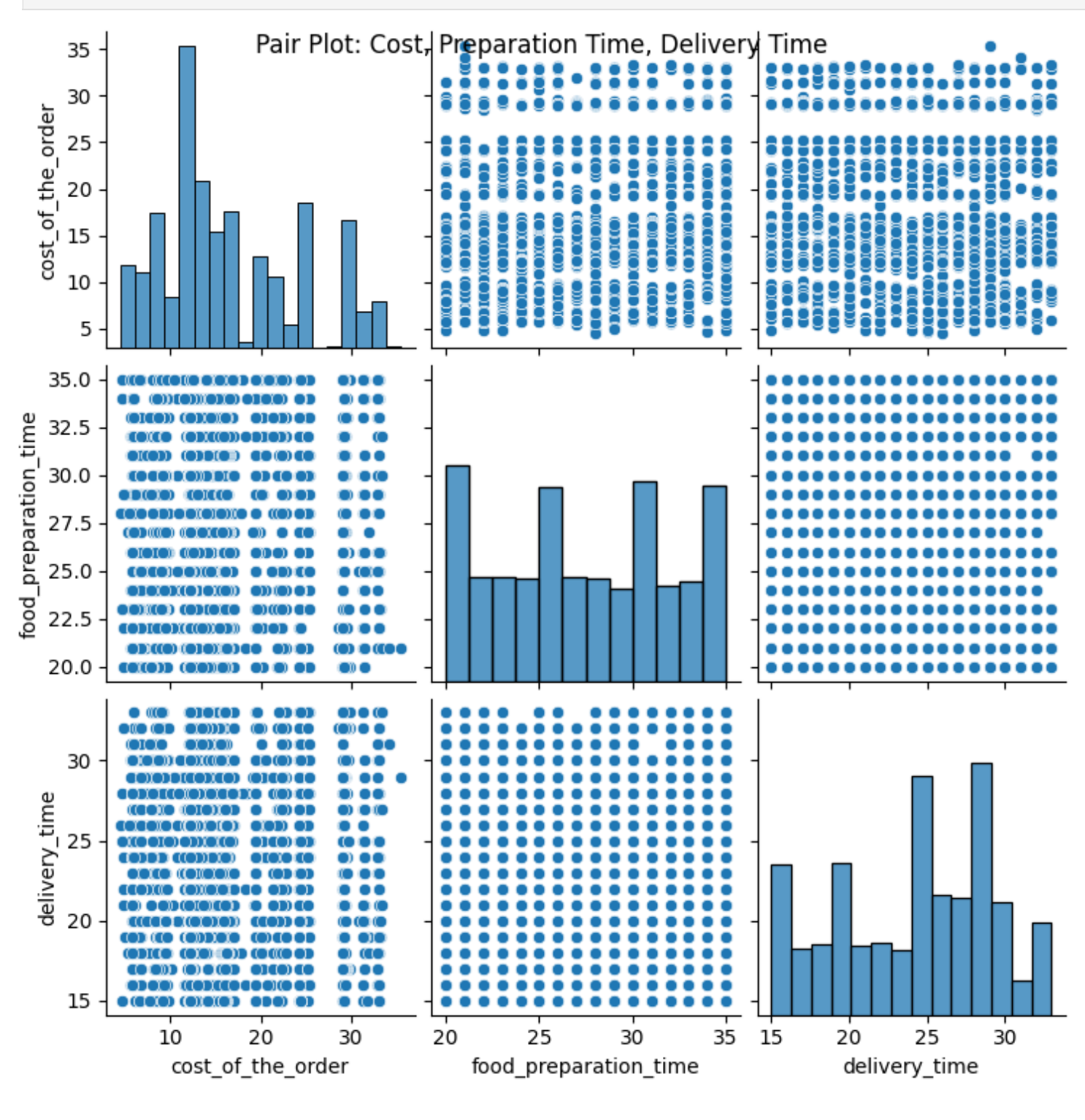


Observations:

From the heat map shown above, there is a low to no correlation between all variables, as most of the variables are close to 0. Cost of the order and food preparation time have slightly positive relation of 0.04, but it is not much. It shows that most of the numerical values are not dependent of each other.

Pair plots of all numerical columns

In [18]:
```
sns.pairplot(df[['cost_of_the_order','food_preparation_time','delivery_time']])
plt.suptitle("Pair Plot: Cost, Preparation Time, Delivery Time")
plt.show()
```



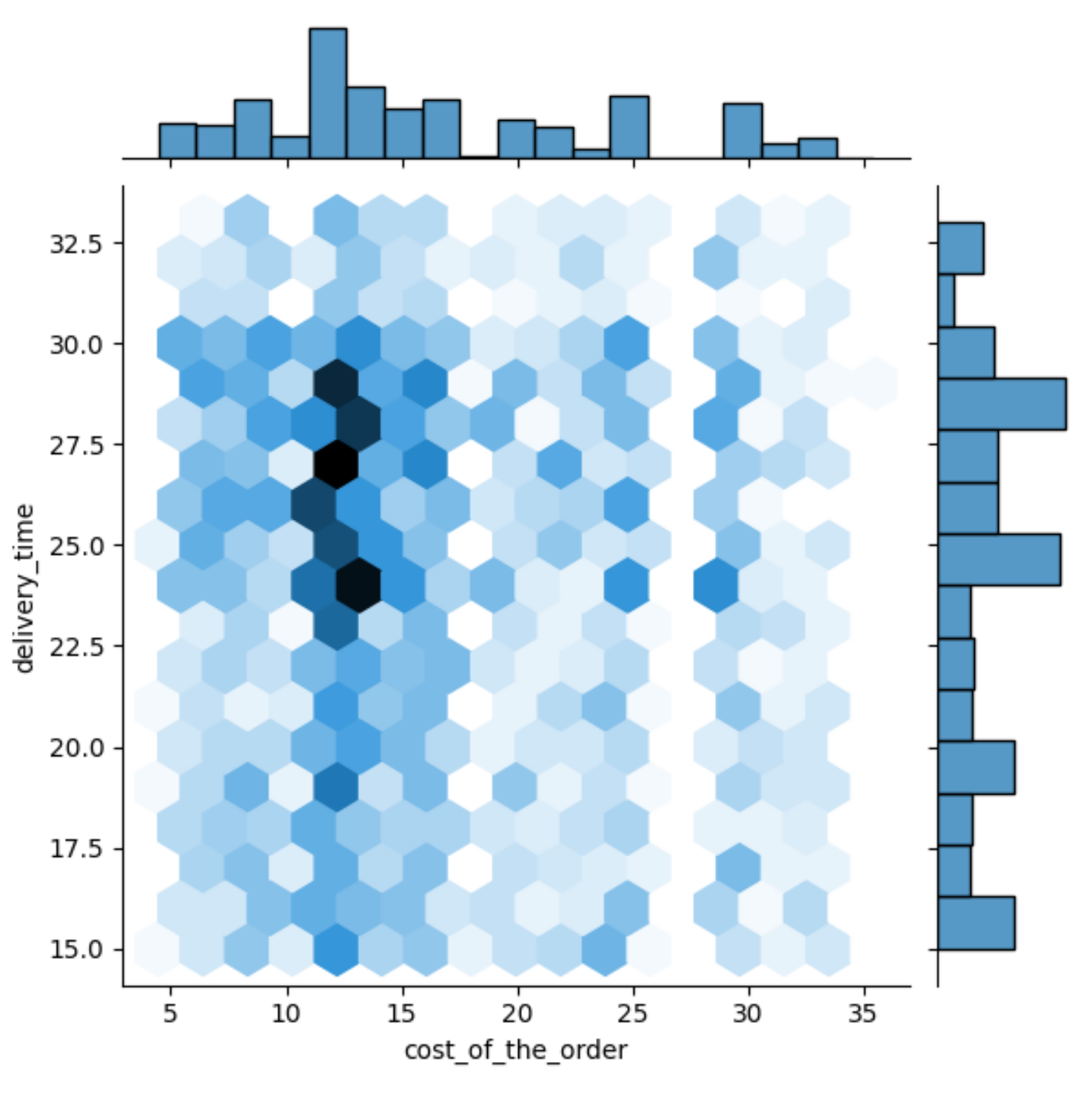
Observations:

From the pair plots above, it shows the cost of the order and food preparation time don't have any correlation, which means the cost of the order is independent and doesn't require more prep time. Even cost of the order and delivery time doesn't have any correlation. Only, food preparation and delivery time are somewhat related with more structured graphs, which means if the food preparation time takes longer than the delivery time also increases.

Joint Plot of Cost of order and Delivery Time

In [30]:
```
plt.figure(figsize =(12, 6));
sns.jointplot(data = df, x = 'cost_of_the_order', y ='delivery_time', kind= 'hex');
plt.show()
```

```
<Figure size 1200x600 with 0 Axes>
```

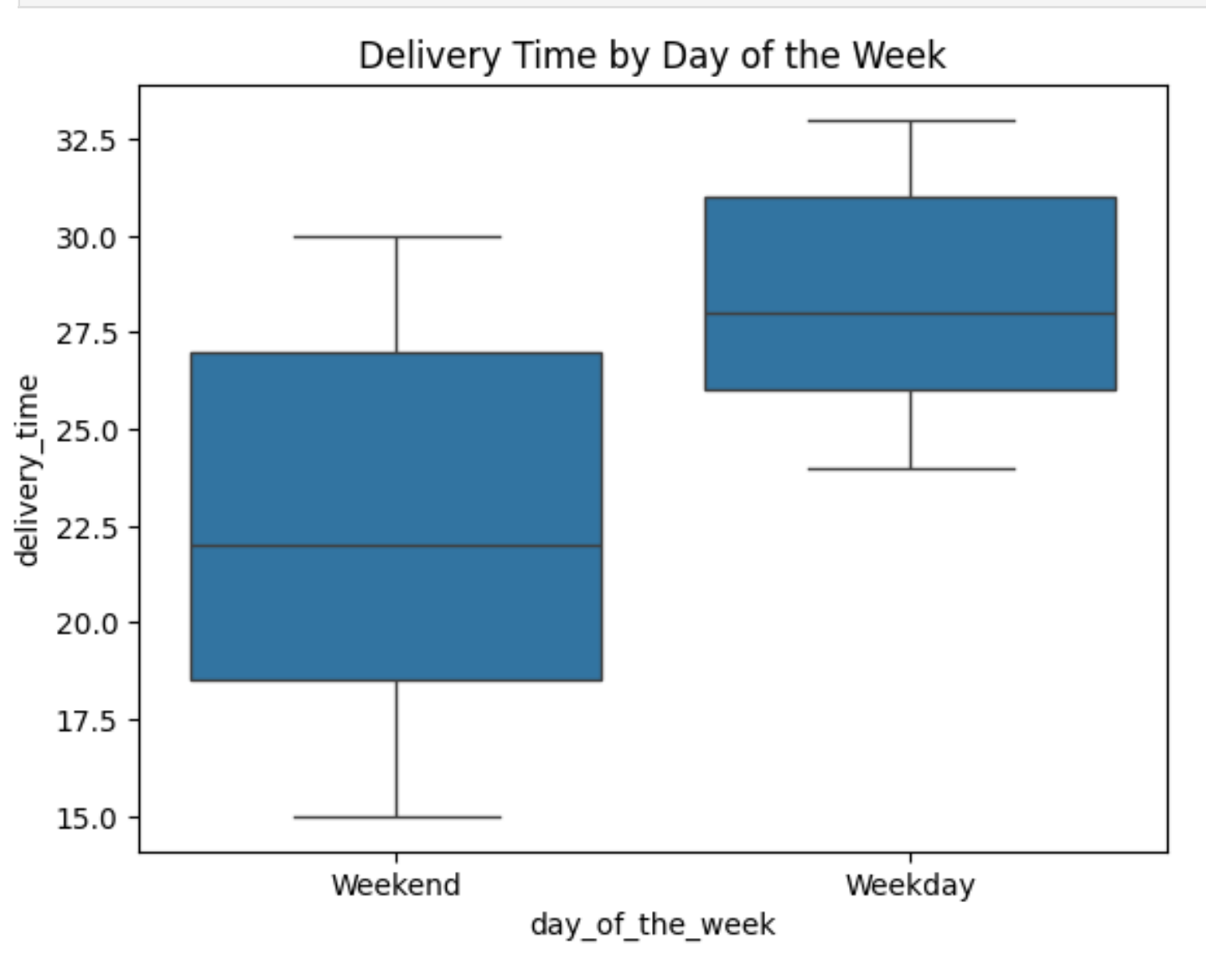


Observations:

From the joint plot above, it shows that customers mostly place orders that cost around 10-15 dollars, which are delivered in around 25-30 minutes. Other than that, it shows that delivery time is fairly stable.

Box Plot of Delivery Time and Day of the Week

In [21]:
```
sns.boxplot(data=df, x='day_of_the_week', y='delivery_time')
plt.title("Delivery Time by Day of the Week")
plt.show()
```

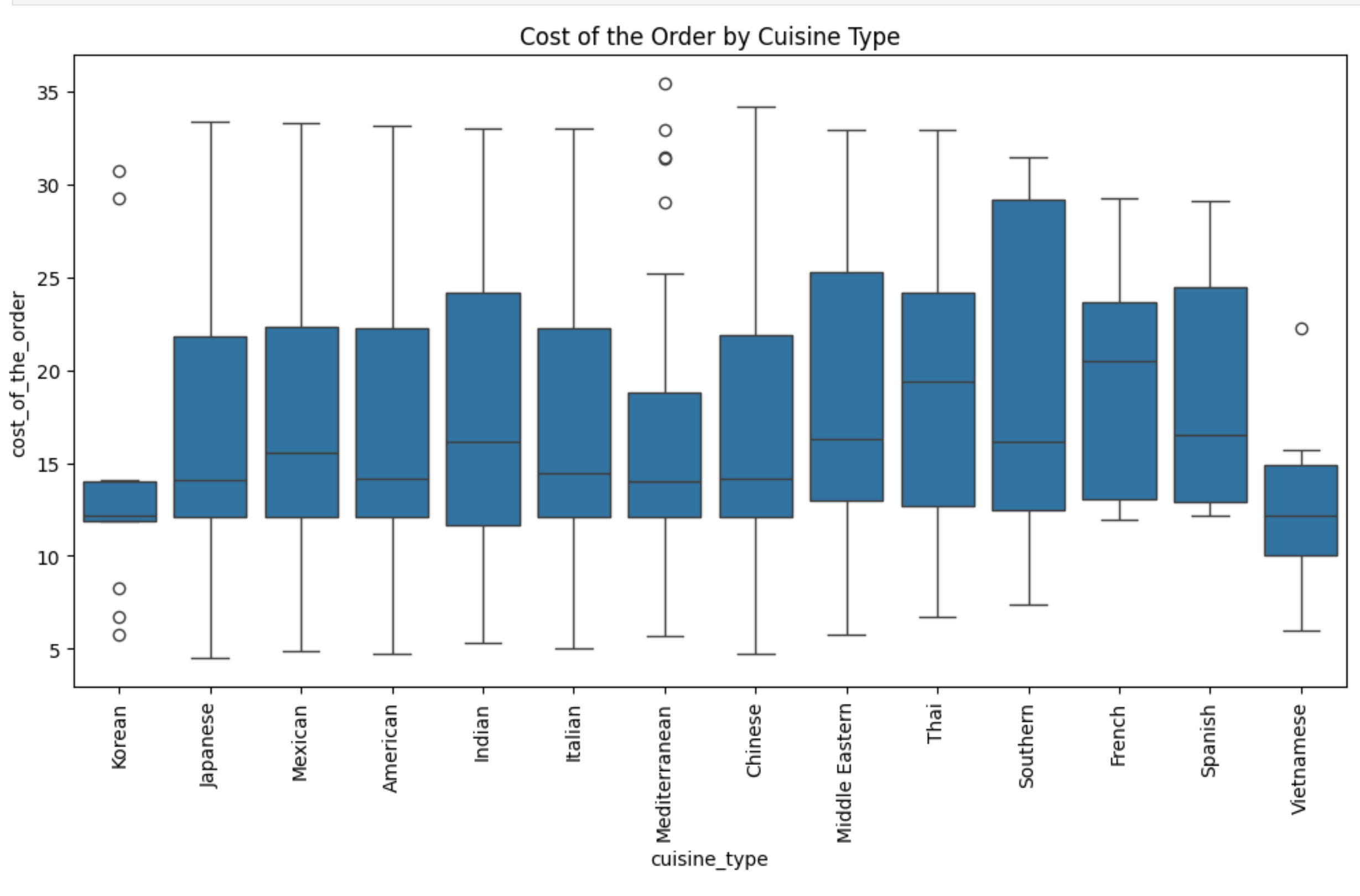


Observations:

From the box plot above, it shows that deliveries take longer on weekdays than on weekends, as the median delivery time on weekdays is around 28 minutes, and median delivery time on weekends is 22 minutes. On weekdays the spread (IQR) is smaller, which means the delivery times are more consistent and on weekends the spread (IQR) is wider, which means there is more variation in delivery times. There are no outliers shown above.

Box Plot of Cost of the order and Cuisine Type

In [22]:
```
plt.figure(figsize=(12,6))
sns.boxplot(data=df, x='cuisine_type', y='cost_of_the_order')
plt.xticks(rotation=90)
plt.title("Cost of the Order by Cuisine Type")
plt.show()
```

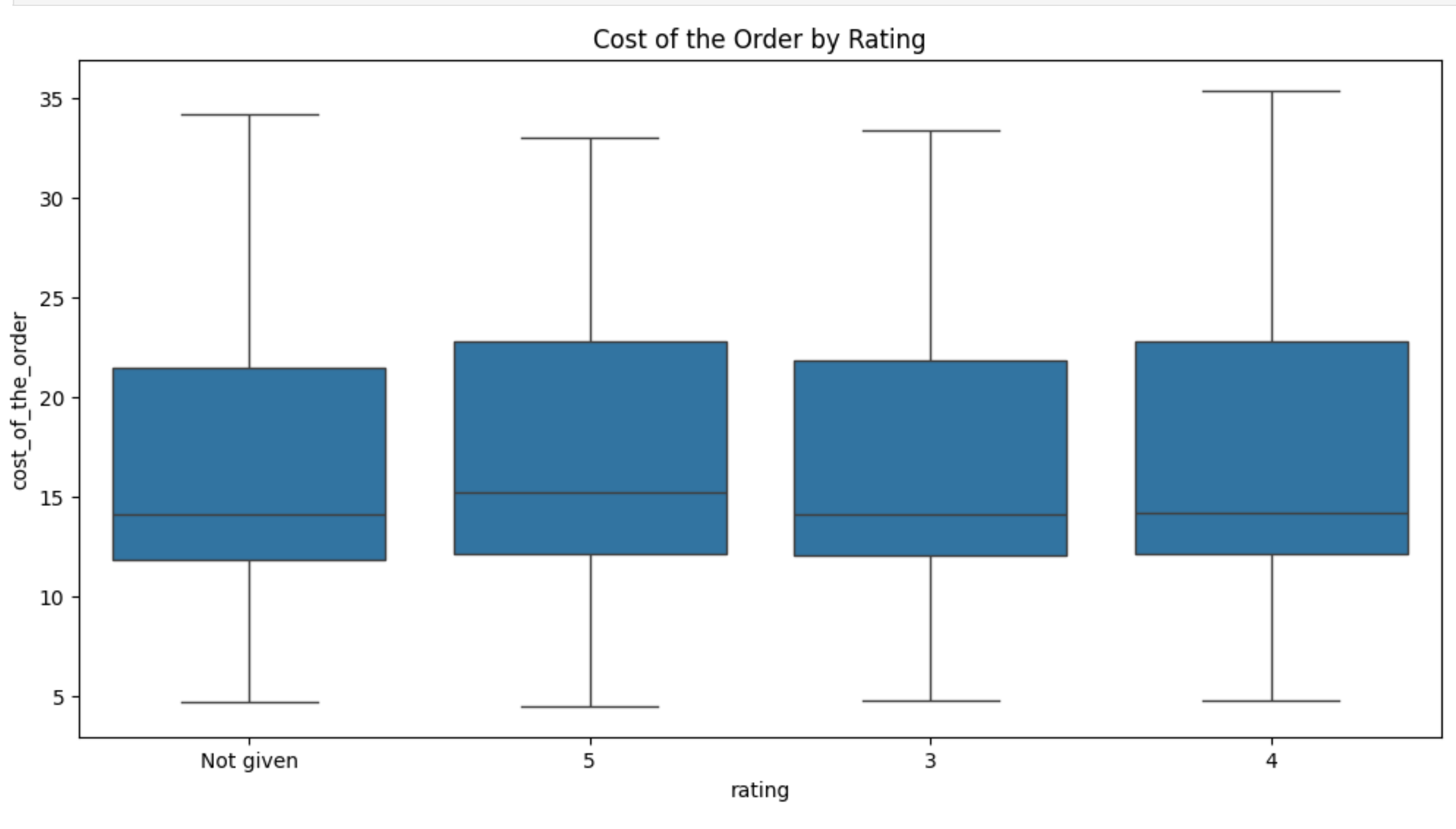


Observations:

From the box plot above, it shows that Southern, French, Thai, and Spanish cuisines orders cost more than the Korean and Vietnamese cuisines as they are the cheapest in price. However, Korean, Mediterranean and Vietnamese cuisines have outliers, which means there are some high cost orders that exist too. Cuisines like Thai, Southern, Spanish, and French their spreads (IQR) are very large, which means their costs vary from cheap to expensive options.

Box Plot of Cost of the order and Rating

In [41]:
```
plt.figure(figsize=(12,6))
sns.boxplot(data=df, x='rating', y='cost_of_the_order')
plt.title("Cost of the Order by Rating")
plt.show()
```

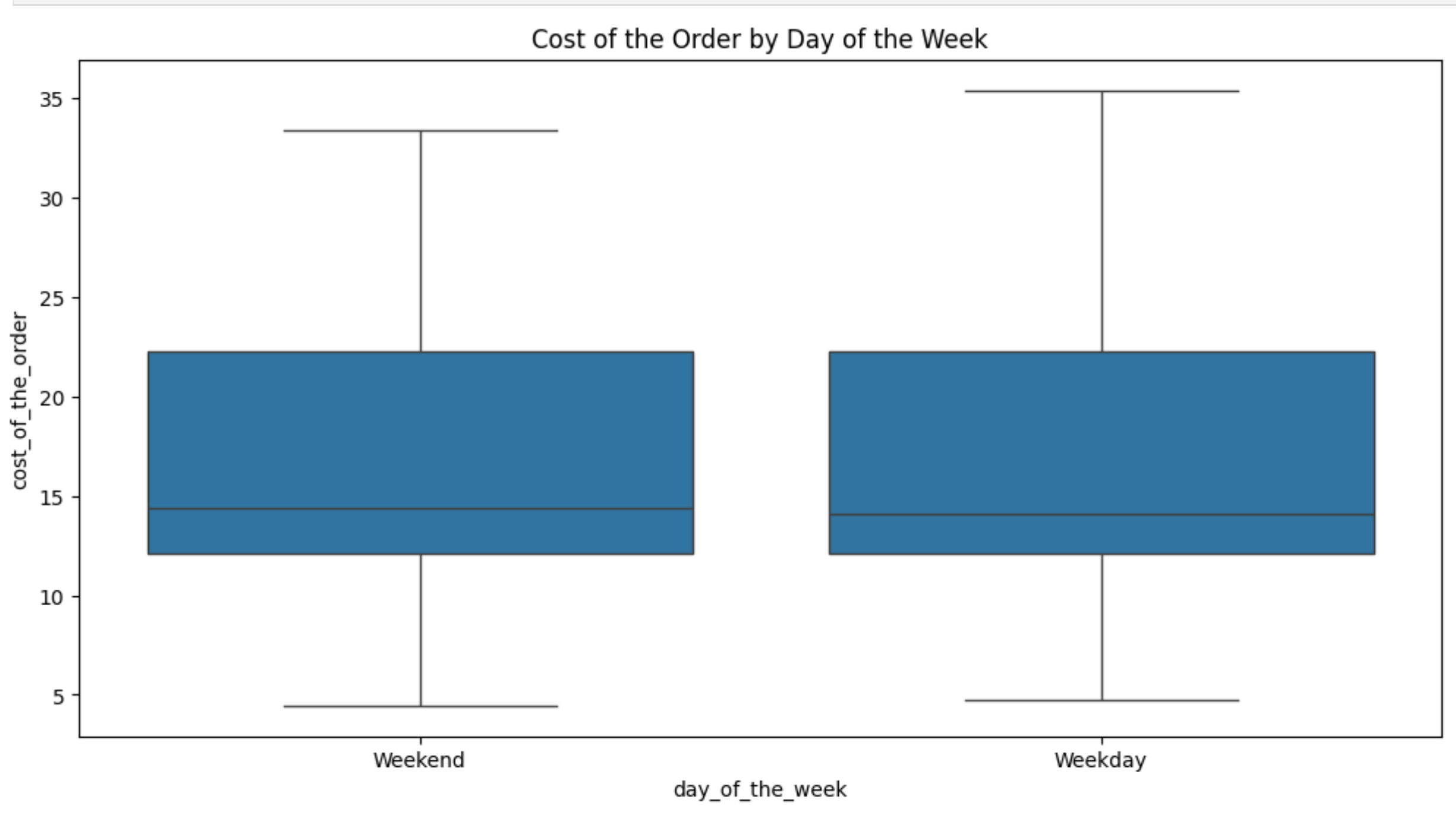


Observations:

From the box plot above, it shows that order cost is not a strong factor in determining the customer ratings. Even if the customer spends less or high, the range of the ratings will be between 3-5.

Box Plot Cost of the Order and Day of the Week

In [43]:
```
plt.figure(figsize=(12,6))
sns.boxplot(data=df, x='day_of_the_week', y='cost_of_the_order')
plt.title("Cost of the Order by Day of the Week")
plt.show()
```

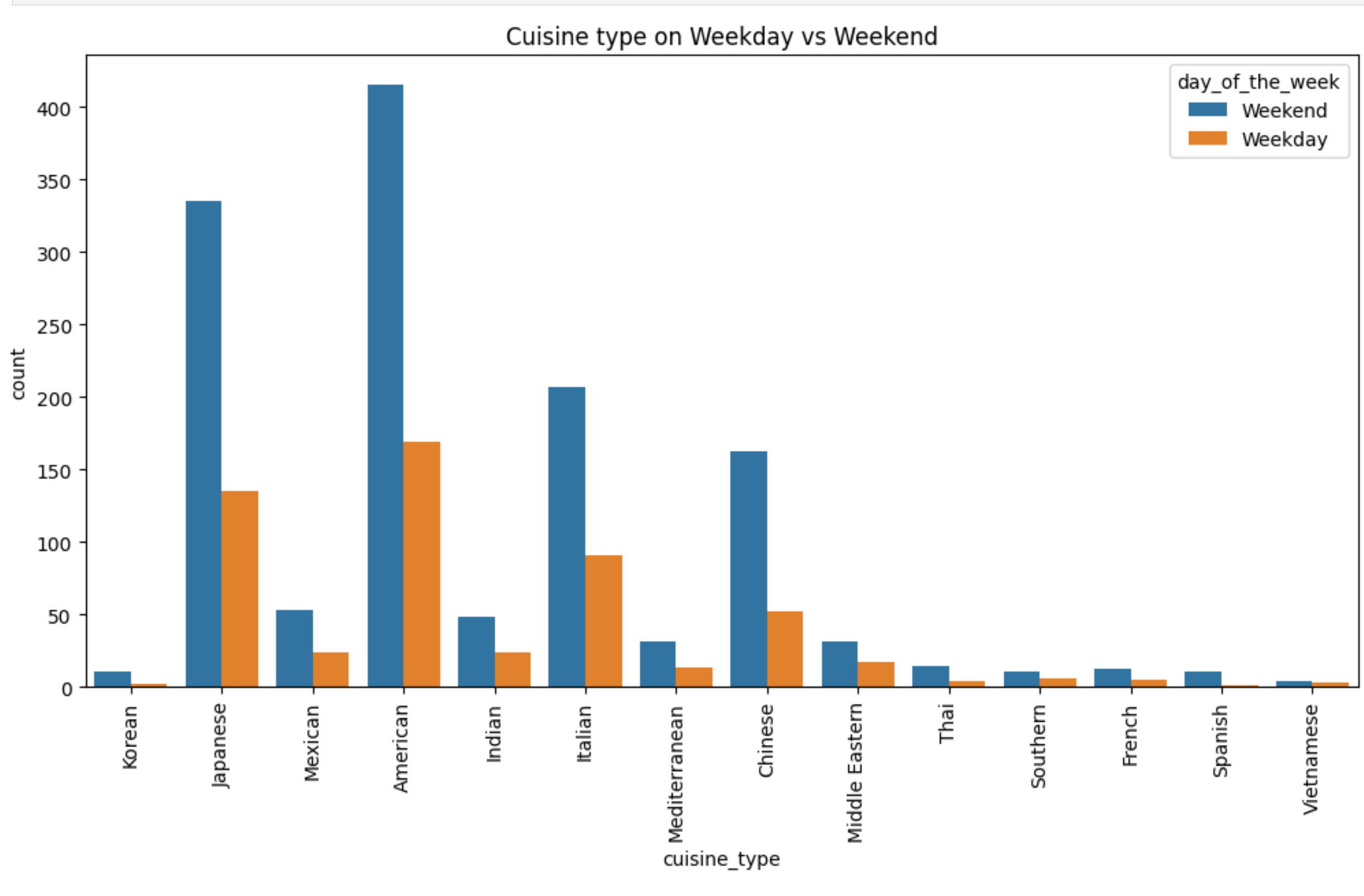


Observations:

From the box plot above, it shows that the median cost of the order is almost the same for weekdays and weekends, which means the customers spend the same amount regardless of the day of the week.

Countplot of Cuisine and Day of the Week

In [23]:
```
plt.figure(figsize=(12,6))
sns.countplot(data=df, x='cuisine_type', hue='day_of_the_week')
plt.xticks(rotation=90)
plt.title("Cuisine type on Weekday vs Weekend")
plt.show()
```



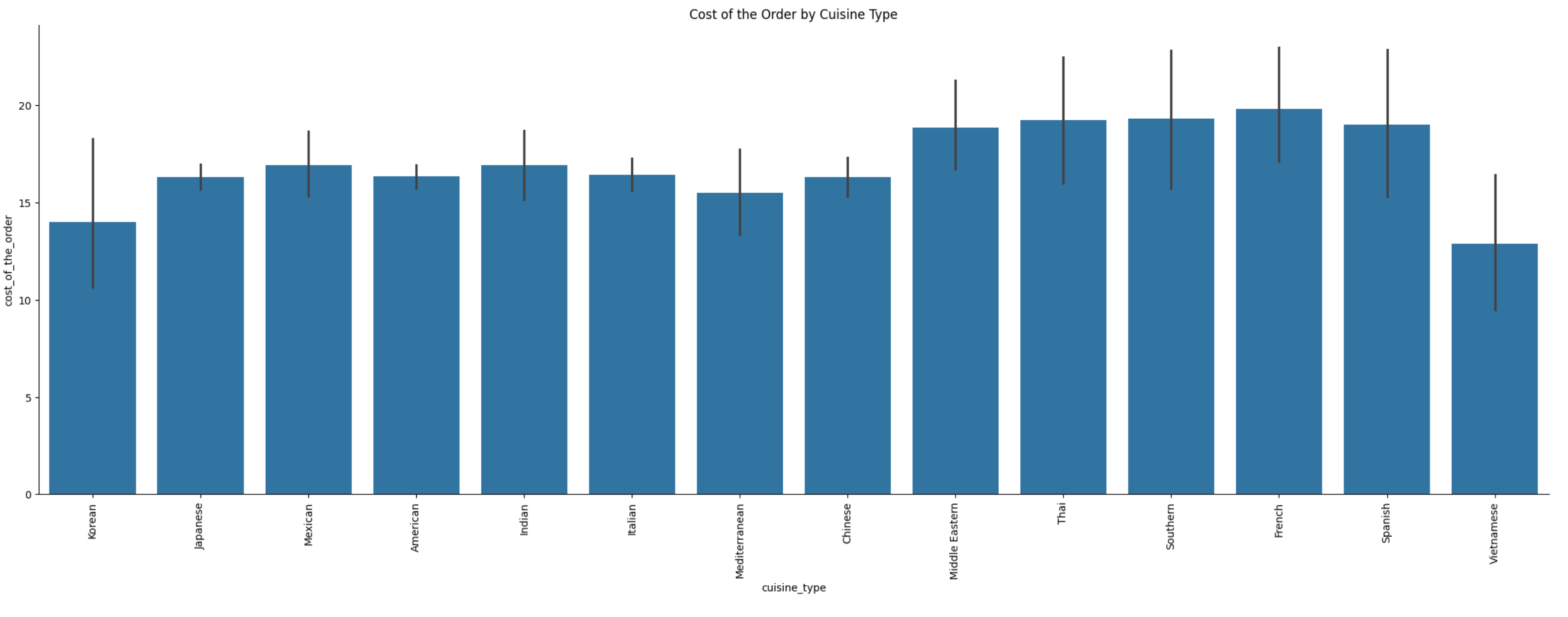
Observations:

From the countplot above, it is clearly shown that people order more food on weekends than weekdays for almost every cuisine. It is shown that most popular cuisines on weekends are American, Japanese, Italian, and Chinese cuisines. Even on the weekdays, people prefer the same cuisines, as American, Japanese, Italian, and Chinese cuisines are higher. Both on weekends and weekdays, Korean, Middle Easter, Thai, Southern, French, Spanish, and Vietnamese, all have low counts of orders.

Cat plot of Cuisine Type and Cost of the order

In [32]:
```
plt.figure(figsize =(15,7));
sns.catplot(data =df, x ='cuisine_type', y ='cost_of_the_order', kind = 'bar', height= 7, aspect = 3);
plt.title('Cost of the Order by Cuisine Type')
plt.xticks(rotation = 90);
plt.show()
```

```
<Figure size 1500x700 with 0 Axes>
```



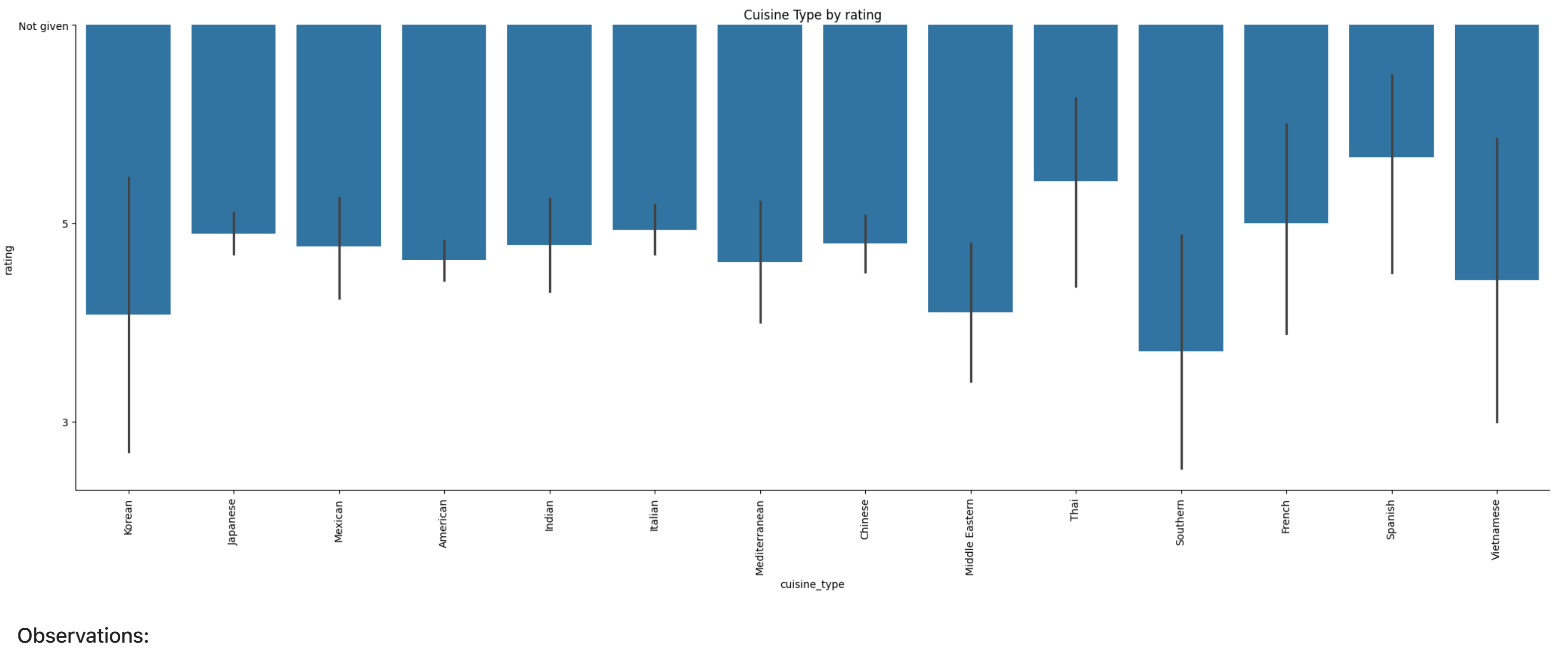
Observations:

From the cat plot above, it shows that the higher cost of cuisines are French, southern, and Spanish cuisines with the highest average cost of about 19-20 dollars. The moderate cuisines are Middle Easters, Thai, Italian, Indian, American, and Mexican cuisines. The lowest cuisines are Korean and Vietnamese cuisines.

Cat Plot of Cuisine Type and Ratings

In [38]:
```
plt.figure(figsize =(12, 6));
sns.catplot(data =df, x ='cuisine_type', y ='rating', kind = 'bar', height= 7, aspect = 3);
plt.title('Cuisine Type by rating')
plt.xticks(rotation = 90);
plt.show()
```

```
<Figure size 1200x600 with 0 Axes>
```

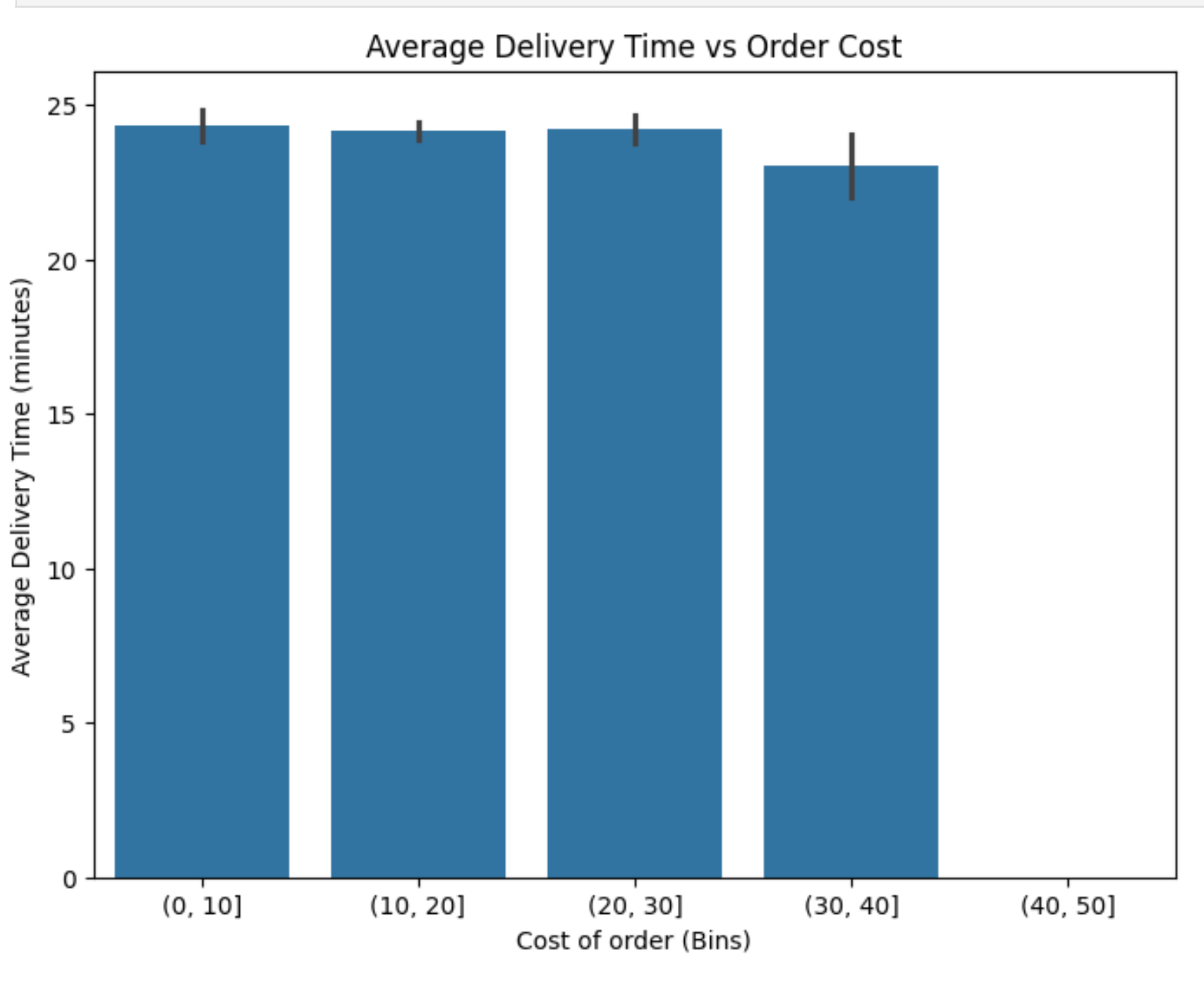


Observations:

From the Cat Plot above, it shows that Japanese, Italian, Spanish and French cuisines have higher average ratings with smaller error bars. Korean, Middle Easter, Southern, and Vietnamese have lower average ratings. Korean, Southern, and Vietnamese cuisines also show large error bars, which means some customers rated them highly and some customers rated them low.

Bar plot of Average delivery time and Cost of the order

In [29]:
```
plt.figure(figsize=(8,6))
df['cost_bin'] = pd.cut(df['cost_of_the_order'], bins=[0,10,20,30,40,50])
sns.barplot(data=df, x='cost_bin', y='delivery_time', estimator='mean')
plt.title("Average Delivery Time vs Order Cost")
plt.xlabel("Cost of order (Bins)")
plt.ylabel("Average Delivery Time (minutes)")
plt.show()
```



Observations:

From the bar plot above, it is shown that most of the delivery times are the same across all bins, which means the order of the cost doesn't affect the delivery time. Error bars are small too, which means delivery times are consistent across all types of orders. The graph shows that delivery time is not strongly depended on order cost.

Question 13: The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer. [3 marks]

In [16]:
```
df_rated = df[df['rating'] != 'Not given'].copy()
df_rated['rating'] = df_rated['rating'].astype(float)

restaurant_ratings = df_rated.groupby('restaurant_name')['rating'].agg(['count', 'mean'])

eligible_restaurants = restaurant_ratings[(restaurant_ratings['count'] > 50) & (restaurant_ratings['mean'] > 4)]

print(eligible_restaurants)
```

```
                            count      mean
restaurant_name
Blue Ribbon Fried Chicken      64  4.328125
Blue Ribbon Sushi              73  4.219178
Shake Shack                   133  4.278195
The Meatball Shop              84  4.511905
```

Observations:

The restuarants that fullfill the criteria to get the promotional offer are Blue Ribbon Fried Chicken, Blue Ribbon Sushi, Shake Shack, and The Meathball Shop.

Question 14: The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Find the net revenue generated by the company across all orders. [3 marks]

In [20]:
```
def commission(cost_of_the_order):
    if cost_of_the_order > 20:
        return cost_of_the_order * 0.25
    elif cost_of_the_order > 5:
        return cost_of_the_order * 0.15
    else:
        return 0

commission = df['cost_of_the_order'].apply(commission)

net_revenue = commission.sum()

print(f"Net revenue: ${net_revenue:.2f}")
```

```
Net revenue: $6166.30
```

Observations:

Net revenue generated by the company across all orders will be 6166.30 dollars.

Question 15: The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed? (The food has to be prepared and then delivered.) [2 marks]

In [6]:
```
total_time = df['food_preparation_time'] + df['delivery_time']

orders_above_60 = (total_time > 60).sum()

percentage_above_60 = (orders_above_60 / len(df)) * 100

print(f"Percentage of orders taking more than 60 minutes: {percentage_above_60:.2f}%")
```

```
Percentage of orders taking more than 60 minutes: 10.54%
```

Observations:

10.54% of orders takes more than 60 minutes to get delivered from the time the order is placed, prepared and delivered.

Question 16: The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends? [2 marks]

In [22]:
```
df.groupby('day_of_the_week')['delivery_time'].mean()
```

Out[22]:

	delivery_time
day_of_the_week	
Weekday	28.340037
Weekend	22.470022

dtype: float64

Observations:

Time it takes to deliver the order on the weekdays is around 28 minutes and time it takes to deliver the order on the weekends is around 22 minutes.

Conclusion and Recommendations

Question 17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business? (You can use cuisine type and feedback ratings to drive your business recommendations.) [6 marks]

Conclusions:

- There are more orders on the weekends than on weekdays, which means customers don't want to cook on the days off and just want to order food.
- American, Japanese, and Italian cuisines are mostly ordered by the customers.
- Most of the orders are less than 20 dollars, however 29.24% of orders are more than 20 dollars, which are on the expensive side.
- Weekday deliveries take around 28 minutes on average, which is slower than the weekend delivers, which takes around 22 minutes on average.
- The Meatball shop restaurant has the highest average rating of 4.51, following with Shake Shack, Blue Ribbon Sushi, and Blue Ribbon Fried Chicken restaurant.

Recommendations:

- The cuisines that drive the most demand on the weekends, like American, Japanese and Italian cuisines, the business should focus more on them with weekend specials.
- Cuisines with less popularity, the business should offer more discounts so people order from those restaurants too, generating more revenue, and this should be done more on the weekends, as their is more traffic then.
- The business should promote highly-rated restaurants on their platform, like The Meatball Shop and Shake Shack.
- The business should encourage customers to leave reviews to get better feedback from the customers.
- As seen from the data, 10.54% of orders take more than 60 mins to deliver, the business should focus on those deliveries and reduce that time for better customer satisfaction.
- The business should come up with a strategy to retain the profitable and frequent customers by offering them free delivery or some points per order.
